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Bachelor of Science, Computer Engineering

- **Programming Languages:** C/C++, Python, Verilog, VHDL, 32-bit MIPS Assembly, Java, HTML, CSS
- **Embedded/Hardware:** Arduino, STM32, ESP32, Xilinx Zynq-7000, CAN/J1939, SPI, I²C, UART, Soldering/Rework, Multimeter, Logic Analyzer, Oscilloscope
- **EDA/Software:** PCB Design (PADS Logic/Layout, KiCad, Altium Designer, Zuken), STM32CubeIDE, Lattice Diamond, SolidWorks, Linux, PetaLinux, Vivado, Vitis, QuestaSim, Quartus Prime, LTspice, Git, Perforce

May 2026 - August 2026

- Designed and shipped a cable harness adapter to maintain pin compatibility for vendor part revision changes
- Architected an automated wire harness tester, integrating CAN bus controllers and a Linux-based display module for verification of machine harness pinouts, further reducing time for setup and testing by nearly 50%
- Managed vendor relations and component procurement, sourcing custom wire harness designs for different machines
- Conducted field testing on a machine to measure and log inrush current for cooling systems and mapped data against interconnect thermal limits to develop a mitigation strategy for overcurrent failures
- Designed the schematic in Altium Designer for a Renesas RA4M1-based system and generated a BOM for cost estimation
- Designed a PCB for a wireless ESP32-based CAN bus device to modify engine RPM during off-machine system testing, eliminating the need for a tethered CAN-to-USB interface

January 2026 - May 2026

- Mentored students in lab with hands-on debugging, instrumentation guidance, and circuit synthesis
- Supported grading homework assignments and quizzes involving analog circuitry

May 2025 - May 2026

- Independently designed the schematic and PCB layout, validated functionality, and assembled an ESP32-based bioimpedance acquisition device utilizing the on-chip DFT of the AD5933 high-precision impedance converter for animal research
- Conducted meetings with stakeholders from the Department of Animal Science to establish design requirements such as built-in SPI SD card data logging and a portable LCD interface and wrote the firmware required to drive both features

May 2024 - December 2024

- Independently developed the schematic and PCB layout for a passive USB Type-C PCB with impedance matching on all differential pairs to support DisplayPort functionality for product testing during assembly
- Revised a serial to fiber optic PCB in PADS Layout and PADS Logic for use during ESD, EMC, and EMI testing
- Reworked prototype PCBs by soldering experimental shielding and 0402-sized bypass capacitors in an effort to minimize capacitive coupling problems during ESD testing and successfully mitigated identified issues

PONG on Custom FPGA Development Board with VGA

- Independently designed the schematic and PCB layout, fabricated, validated, and programmed a custom FPGA board based around Lattice Semiconductor's MachXO2 7000HC FPGA, featuring onboard power regulation, configurable I/O voltage banks, synchronous 640x480 VGA timing logic, a custom FTDI FT232RL-based JTAG programmer, and expansion headers
- Implemented a two-player PONG game in VHDL (see <https://pianetto.net/projects.html> for more details)

- Developed a custom memory-mapped state-machine IP block for PWM and 4-phase motor control in VHDL, enabling MIDI playback through floppy drive motors using a Xilinx Zedboard running Linux
- Reverse-engineered a flatbed scanner PCB by analyzing datasheets of ICs and modifying existing circuitry to directly control the 4-phase motor from the Zedboard's PMODs using onboard Darlington transistor arrays and external logic level shifters
- Rewrote and optimized C++ drivers to remove dependencies on large libraries, successfully shrinking the software footprint from 550 KB to under 250 KB to fit the FSBL with an I²C/SPI-based MIPI camera within the limited OCM of the Zedboard